Methods: eso_137-001

ALMA Thesis Planner (automated pipeline)

Step 1: Data retrieval and preparation

Download the pipeline-calibrated measurement set and QA2 image/cube products for member OUS 'uid://A001/X196/X78' (project 2013.1.01364.S) directly from the ALMA Science Archive. Verify the delivered products against the QA2 report and weblog: confirm the frequency setup (98.6–100.5, 100.6–102.5, 110.8–112.7, and 113.3–113.8 GHz), confirm which spectral window contains the redshifted CO(1-0) line, and record the exact observed frequency of the line as stated in the QA2 documentation for this dataset. Do not assume a redshift or observed-frequency value that isn't explicitly given in the delivered documentation — the rest frequency of CO(1-0) (115.27 GHz) only tells you the line falls somewhere in this tuning once shifted by the cluster's recession velocity, and that precise observed value must be confirmed from the QA2/imaging metadata, not assumed. Load the calibrated cube into CASA ('importfits'/'imhead'/'viewer' as appropriate) at its native angular resolution (7.892") and velocity resolution (0.643 km/s).

Step 2: Mosaic footprint and coverage check

Using 'imhead', the primary-beam response image, and the pixel-value mask stored with the QA2 cube, map out the actual usable field of view and sensitivity pattern of the delivered mosaic. Overlay this footprint on existing multiwavelength imagery of ESO 137-001 (e.g., archival X-ray and Hα tail morphology, if the student already has access to such images independent of this proposal) purely to orient the CO map spatially — not to assume the ALMA mosaic reaches the full 80 kpc (X-ray) or 40 kpc (APEX single-dish CO) extents quoted in the proposal abstract. Establish, from the footprint and the noise map alone, how far down the tail (in kpc, converting from arcsec using the cluster's known distance) the ALMA data actually provide usable sensitivity. All subsequent analysis is bounded to this confirmed extent.

Step 3: Moment map generation

Within CASA, use 'immoments' to produce moment 0 (integrated intensity), moment 1 (intensity-weighted velocity), and moment 2 (velocity dispersion) maps at native resolution, applying a conservative masking threshold (e.g., a smoothed/masked cube approach: smooth to ~2× beam, mask at a fixed S/N threshold such as 3–5σ, then apply that mask to the native-resolution cube before computing moments) to suppress noise in low-S/N regions of the tail. Separately generate moment maps restricted to the stellar-disk region and to the tail region as defined by the footprint check in Step 2.

Step 4: Missing-flux / maximum-recoverable-scale diagnostic

Because the observations table lists only a single 12-m array member OUS with no ACA or total-power component, quantify the risk of resolved-out diffuse flux directly from the delivered data rather than by reference to any external total-flux measurement. Determine the shortest baseline length used in the calibrated measurement set (via CASA's 'plotms'/'listobs' or the imaging weblog) and compute the corresponding maximum recoverable angular scale (MRS), or read the MRS reported directly in the QA2 documentation for this imaging run. Convert the angular sizes of the tail structures under study (as identified in Steps 2–5) from kpc to arcsec at the cluster distance, and compare these directly to the MRS. Report explicitly, as a qualitative but quantified caveat, whether tail structures of interest approach or exceed the MRS, flagging any resulting risk to gas-mass or morphology measurements. Do not import or cite any specific external flux value (e.g., an APEX total flux) in this comparison, since none appears in the data description or in any archive-linked publication for this project.

Step 5: Clump identification

Apply a standard 2D or 3D clump-finding method to the moment-0 map (or to the full cube, if a 3D approach such as a dendrogram is used) — e.g., the ‘astrodendro‘ package for a dendrogram-based decomposition, or a simple SExtractor-style connected-component thresholding on moment 0 — separately in the disk region and the tail region, using a consistent detection threshold (e.g., 3σ) across both regions for a fair comparison. Before proceeding, apply the pre-defined minimum-deliverable test: count the number of independent resolution elements (beam-sized apertures) along the confirmed tail extent that reach S/N > 3 in integrated intensity.

- If $\geq$ ~5 independent resolution elements meet this threshold, proceed to individual clump-by-clump kinematic analysis (Step 6).
- If fewer than ~5 do so, abandon the individual-clump approach for the tail and instead extract a single tail-integrated spectrum from an aperture (or a small number of large bins) covering the confirmed tail extent, and report an upper limit on integrated CO flux (and hence molecular gas mass) and on velocity dispersion, explicitly labeled as a non-detection result with quantified limits.

Step 6: Per-clump (or per-bin) measurements

For each detected clump, or for the tail-integrated bin(s) per the branch taken in Step 5, measure: (a) centroid velocity, from a Gaussian or moment-based fit to the extracted spectrum; (b) integrated CO(1-0) flux, converted to molecular gas mass using a CO-to-H2 conversion factor (α_CO); and (c) velocity dispersion, from the line width of the extracted spectrum (or from the local moment-2 value). Explicitly separate statistical uncertainty (from noise/fit uncertainty) from the systematic uncertainty introduced by the choice of α_CO. State clearly that the standard disk value of α_CO may not apply to tail gas, since ram-pressure-stripped and possibly in-situ-formed material may differ in metallicity and excitation from disk gas, and that α_CO for such stripped/tail material is not settled in the literature. As a sensitivity check, recompute tail gas masses under 2–3 literature-motivated α_CO choices and present the resulting range alongside (not blended into) the statistical uncertainty, rather than adopting a single value as if it were well established.

Step 7: Kinematic comparison

Compare the spatial distribution of centroid velocities and velocity dispersions between disk clumps and tail clumps (or the tail-integrated bin, in the non-detection branch), using the moment 1 and moment 2 maps from Step 3 alongside the per-clump measurements from Step 6. Characterize whether the tail velocity field shows a smooth, monotonic gradient consistent with ballistic stripping directly off the disk, or whether it shows patchy/decoupled velocity structure and locally enhanced dispersion more consistent with gas that has locally condensed within a turbulent, mixed ISM–ICM medium. Given the limited number of independent resolution elements expected along the tail, present this comparison as a descriptive characterization of the kinematic structure (or, in the non-detection branch, as bounded upper limits) rather than as a statistically powered test of two populations. Summarize the result together with the Step 4 missing-flux diagnostic and the Step 6 α_CO systematic, so that the final kinematic characterization is presented with its governing systematic and sensitivity caveats attached throughout, not appended as an afterthought.